[Lecture Note] Measurement Errors, Reliability, Validity

GE2021 Research Technology, Spring 2025

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Mon, 26 May 2025

Learning Objectives

- 5-1 Recognize measurement errors and describe how to categorize them.
- 5-2 Compare and contrast the interrater reliability, test-retest reliability, and internal consistency reliability.
- 5-3 Analyze the different types of validity: face validity, content validity, construct validity, criterion validity, concurrent validity, and predictive validity.

SHORT VIDEO INTRODUCTION

https://www.youtube.com/watch?v=oE71Onl5K-s

Defining Measurement Error

- Measurement error: when the data collected does not represent reality because of the way it was measured.
- Exclusive to quantitative design

Types of Measurement Errors

- Random Error: when the measurement error in the collected data is inconsistent and has no pattern.
  - When measurement error is small and random, there still may be high-quality data available.
- Systematic Error: when data collected indicates that the measure is not accurately measuring the concept.
  - Far more problematic than random error

  - Can be related to the measurement being used, the way the data is being collected, or other environmental factors unrelated to the measurement.

Reliability

- Reliability in quantitative design: the consistency in a measurement; data collection is reliable if it yields the same results when used on different subjects, populations, or settings.
- Reliability in qualitative design: the study's replicability, dependability, and consistency.
- Accuracy: participants' answers about a topic and whether or not the answers are free from error.
- Reliability does not mean accuracy.

Inter-Observer or Inter-Rater Reliability

- There is more than one researcher present in data collection, and all observers keep separate ratings of the data to then compare responses to determine a degree of consistency.
- More common in qualitative studies

Test-Retest Reliability

- Associated exclusively with quantitative design
- Ensures data collection tool consistently yields the same results regardless of the passing of time.

Internal Consistency Reliability

- Often used in surveys and questionnaires and associated with quantitative designs
- Including multiple questions that measure the same variable to make sure the participant's answer was not due to error.

Validity

- Validity: the ability or potential of a data collection tool to capture and measure the construct or phenomenon that the researcher is interested in measuring.
- In quantitative design, it is tightly related to the measurement of a concept, whereas in qualitative design it relates to how a concept is being captured through texts, images, or sounds.

Face Validity

- When researchers decide to classify the measurement instrument or tool as an accurate measurement of the construct in question.
- A subjective assessment of the validity of a measurement instrument
- Does not imply the measurement is valid, but implies that for the sake of the researcher's discussion, the measurement is considered valid.
- Often used because measurement tools cannot always be confirmed as valid

Content Validity

- How well a measurement tool captures all aspects of the construct that is being tested or measured.

Construct Validity

- How well a measurement is able to truly measure the construct in question; how accurately researchers have operationalized a construct.

Criterion Validity, Concurrent Validity, and Predictive Validity

- Criterion validity: for cases when researchers are no satisfied with the way the construct has been measured and decide to create a new measurement for a construct that has already been measured.
- Criterion validity needs to establish both concurrent validity and predictive validity.
- Concurrent validity: occurs when comparing a new test with an established measurement tool and examining the matching results
- Predictive validity: occurs when a measurement test produces the same results over time.
- Difference Between Reliability and Validity of the Measurement Instrument

Class Activities

- Divide the classroom into groups of two to three students. Ask each group to create a survey tool with six to eight questions to measure stress among college students. Give each group, 10 min to complete this exercise. At the end of 10 min, collect each tool and distribute it to a different group. Ask each group to critique their classmate's stress tool specifically commenting on the face validity of the tool. To what extent does the tool comprehensively capture the construct of stress among college students? What may be missing from each tool?

- You are working with a group of community partners to measure perceived social support among low-income mothers in Detroit, Michigan. Your partners are interested in working with you to create a survey tool to be distributed to these mothers. Work with a group of two to three students to discuss your process for designing this tool. Outline the steps you will take to do this. For example, will you immediately begin to develop questions, or will you first look online and in the literature for existing instruments? If you find an instrument online what will you do with it? If you propose to create your own questions, how will you go about ensuring the questions are valid and reliable?

- Divide the classroom into groups of four to five students. Assign each student group to go online and find a peer-reviewed article that discusses the development of a measurement tool. Ask each group to review this article and prepare a short presentation for the class, outlining the measurement tool and describing the development of the tool, including any assessment of reliability and validity conducted by the researchers.

House Keeping

Announcement

Assignment

Assignment1: Assignment Title

- **What to do**
  - A
  - B
- **Format and Requirement**
  - PDF format, < ?? pages
  - file name should be include your student id and name
    - stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
  - APA, Double-Space, No title page, 12 points, Reference section if necessary
  - Chinese name in English, English name, Student ID, Class name and section (e.g. GE2021, W09; MGS3001, W01)
- due date
  - by Date(DAY) 11:59PM
  - NO LATE SUBMISSION ALLOWED!!!!

Reference